Ternary Sufficient-Statistic Import for State-Space Language Model Training

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Abstract

This paper introduces Ternary Sufficient-Statistic Import (TSSI), a training object for state-space language models with ternary weights. The motivating question is deliberately narrow: before iterative optimization begins, can a corpus be compiled into model-native $\{-1, 0, +1\}$ commitments that already answer part of the cross-entropy descent problem? TSSI does not import text, examples, retrieval memory, or a teacher model. It streams the corpus once, accumulates original cross-entropy directional moments in a fixed SSM feature map, realizes stable directions as sparse ternary sign/zero commitments, and then leaves the remaining error to ordinary residual training on the same next-token objective. Initial gates on byte-level v12 shards show that corpus-derived ternary commitments beat density-matched random and sign-shuffled controls, including over live Vulkan SSM hidden states and record-disjoint held-out trace rotations. The result supports TSSI as a training-theoretic mechanism: useful corpus information can be converted into a ternary SSM artifact without becoming raw memorization or a replacement objective.

1 Introduction

The usual framing of language-model training treats the corpus as something that must be visited many times by an optimizer until useful structure has been written into the parameters. For continuous weights this is natural: the model must discover both a direction and a magnitude in a high-dimensional real space. For a ternary state-space model, however, the first useful question is smaller and more discrete. A coordinate often only needs an initial answer to one of three possibilities: negative, zero, or positive. If the corpus already contains enough directional evidence for some of those answers, then forcing all of that evidence to pass through ordinary training steps is not a law of learning; it is a design choice.

The phrase "dataset import" is easy to misread. A literal import of a dataset into a model is not the object studied here. Storing examples, building a lookup table, or hiding retrieval behind the parameter interface would evade the training problem rather than improve it. The admissible object is narrower: a sufficient statistic for a particular training objective. In this paper the objective remains next-token cross-entropy, and the imported artifact is only a set of sparse ternary commitments in model coordinates. Anything not supported by stable corpus evidence is left as zero, so the ordinary residual training process still owns the unsolved part of the model.

This gives a different way to ask what a corpus can do before training. Instead of asking whether text can be compressed into weights, TSSI asks whether cross-entropy moments can be compiled into the same ternary decision surface that training would otherwise have to reach by repeated updates. The compiler streams contexts and targets, accumulates directional moments under the original loss, applies an abstention threshold, and writes only $\{-1, 0, +1\}$ decisions. The zero state is important: it is not failure, but an explicit refusal to pretend that weak evidence is already knowledge.

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The empirical claim is correspondingly scoped. On byte-level v12 shards, a bigram feature gate, residual cross-entropy pilots, a deterministic recurrent `ssm_state` proxy, live Vulkan SSM hidden-state traces, and record-disjoint held-out rotations all show the same qualitative behavior: corpus-derived ternary commitments carry real predictive signal beyond density-matched random or sign-shuffled controls. The live SSM result matters most because it places the statistic in actual state-space coordinates rather than in a purely symbolic proxy. The record-disjoint rotations matter because they reduce the simplest memorization explanation: the held-out records change, yet the all-layer imported commitments remain better than matched random commitments.

The contribution is therefore not a new dataset, not a teacher-distillation method, and not an associative memory. It is a training-theoretic object for ternary SSMs: a corpus-to-commitment map that preserves the original cross-entropy objective and initializes part of the ternary search problem before residual optimization. The rest of the paper defines this object, separates it from nearby prior art, states the validation protocol, and reports the current gates.

2 Background and Related Work

Dataset distillation and condensation. Dataset distillation asks for a small synthetic training set that can replace a larger corpus for downstream optimization [Wang et al., 2018]. Text-level variants such as DiLM move this idea into language by generating synthetic text that trains a model efficiently [Maekawa et al., 2024]. TSSI is not a synthetic dataset. It has no generated examples and no second corpus that the optimizer later consumes. The output is a model-coordinate artifact: a sparse ternary commitment pattern derived from the original cross-entropy moments. The distinction matters because a distilled dataset still delegates learning to future optimization over examples, while TSSI answers a bounded subset of the ternary parameter decisions directly.

Data-dependent initialization. Another nearby line uses data to choose a better initialization for continuous training. Classical data-dependent initializations match early activation or gradient scales [Krähenbühl et al., 2015]; GradInit tunes initialization to improve early optimization dynamics [Zhu et al., 2021]; AutoInit searches for initialization rules that stabilize neural networks automatically [Bingham and Miikkulainen, 2021]. These methods remain continuous in spirit. Their goal is to place the optimizer in a numerically favorable region. TSSI instead uses data to make discrete ternary commitments. The imported state is not a scale, variance, or conditioning adjustment; it is a partial answer to the model's sign/zero decision problem under the original loss.

Ternary and low-bit language models. Low-bit language-model work shows that large networks can preserve useful behavior under severe quantization [Wang et al., 2023, Ma et al., 2024]. Earlier ternary-weight methods also study how to train or quantize networks with $\{-1, 0, +1\}$ weights [Li et al., 2016]. Those results make the present question plausible, but they do not define a corpus-to-ternary import object. The usual quantization problem starts with trained or trainable real-valued parameters and asks how to represent them cheaply. TSSI starts one step earlier: given a corpus and a fixed SSM feature map, which ternary coordinates already have enough evidence to be committed before residual training continues?

State-space coordinates. The coordinate system is also part of the claim. Modern SSMs such as Mamba derive much of their sequence behavior from recurrent hidden states rather than only from local token features [Gu and Dao, 2024]. A corpus statistic that passes in a bag-of-symbols proxy may still fail when placed into live recurrent state coordinates. For that reason, the strongest current TSSI evidence is not the byte/bigram gate by itself, but the live Vulkan SSM hidden-state trace and the record-disjoint rotations over that trace. The artifact is intended to be model-native, not merely corpus-native.

Memory and teacher boundaries. TSSI should also be separated from retrieval, associative memory, and teacher-based distillation. A retrieval system keeps examples available at inference time. A teacher system transfers behavior from a stronger external model. TSSI does neither. The compiler sees only corpus tokens and fixed SSM feature coordinates, writes ternary sign/zero commitments, and then disappears. After import, the remaining burden is still optimized by the same original cross-entropy objective. This residual contract is the main semantic guardrail: if an import path cannot preserve it, it is not TSSI as defined in this paper.

3 Ternary Sufficient-Statistic Import

TSSI is a corpus compiler for a fixed model-coordinate view of the training objective. It should not be read as an attempt to solve the model in closed form. The compiler only asks whether the corpus gives enough first-order evidence to commit some ternary coordinates before the residual optimizer takes over.

3.1 Corpus Moment Accumulation

Let $D = \{(c_t, y_t)\}_{t=1}^{T}$ denote a stream of next-token contexts and targets. Let $\psi(c_t) \in \mathbb{R}^m$ be a fixed feature map in the coordinate system where the import will be written. Depending on the gate, ψ can be a byte bigram feature, a deterministic recurrent proxy, or a live SSM hidden-state summary. Let $p_0(\cdot \mid c_t)$ be the reference model distribution before import. The cross-entropy gradient for a linearized readout coordinate has the form

$$(p_0(\cdot \mid c_t) - e_{y_t}) \otimes \psi(c_t), \tag{1}$$

where p_0 is the current reference distribution and e_{y_t} is the one-hot target. TSSI accumulates this quantity over the corpus:

$$G_D = \sum_{t=1}^{T} (p_0(\cdot \mid c_t) - e_{y_t}) \otimes \psi(c_t). \tag{2}$$

For the fixed feature map and reference distribution, G_D is the sufficient statistic used by the import rule. It is not a sufficient statistic for all future training; it is sufficient only for the bounded question asked here: which ternary coordinates have stable first-order CE evidence?

3.2 Ternary Realization

The imported artifact θ_I is a ternary sign/zero commitment. Since $G_D[j]$ is a gradient coordinate, a positive value points toward decreasing the parameter and a negative value points toward increasing it. The ternary import therefore commits in the descent direction:

$$\theta_I[j] = \begin{cases} -1 & \text{if } G_D[j] > \tau_j, \\ 0 & \text{if } |G_D[j]| \le \tau_j, \\ +1 & \text{if } G_D[j] < -\tau_j. \end{cases} \tag{3}$$

The thresholds τ_j implement abstention. They can be set by a target density, by calibration data, or by a coordinate-wise confidence rule. The important property is semantic, not cosmetic: zero means "do not claim this coordinate yet." It keeps uncertain coordinates available for residual training instead of converting weak corpus evidence into wrong signs.

3.3 Why This Is an Import

The imported object is smaller than a dataset and weaker than a trained model. It contains no examples and no token-addressable memory. It also does not contain real-valued magnitudes except through the choice of which ternary cells are committed and which are left at zero. This is the reason the mechanism is natural for ternary SSMs. The compiler is not trying to discover a full continuous solution; it is trying to answer the first discrete layer of the training problem.

3.4 Residual Original-CE Contract

After import, the model is still trained on the original objective. Let δ denote the remaining trainable degrees of freedom, including uncommitted ternary coordinates and any residual capacity. The post-import training problem is

$$\min_{\delta} \ \mathrm{CE}_D(\mathrm{realize}(\theta_I, \delta)). \tag{4}$$

This contract is the guardrail that makes TSSI a training mechanism. If the import writes helpful signs, residual training begins from a better ternary artifact. If the import writes harmful signs, the original CE objective exposes the harm. No alternative loss, teacher target, or retrieval path is introduced.

3.5 Failure Modes

The definition deliberately leaves room for failure. The feature map may be too weak, the corpus may not provide stable signs, the thresholds may admit harmful coordinates, or the imported coordinates may help a proxy but fail in live SSM state space. These are empirical and mathematical failure modes, not implementation details. The validation protocol in the next section is built to attack them directly.

4 Validation Protocol

The validation target is not to show that an imported artifact looks good on the same examples that created it. The target is narrower and stricter: a TSSI artifact must improve held-out cross-entropy or held-out CE-gradient sign alignment in a way that cannot be explained by density, sign balance, logit scale, or example leakage.

4.1 Identifying Checks

The first identifying checks compare the imported artifact θ_I against a reference start θ_{ref} and density-matched random ternary artifacts. The import is identifying only if

$$A_{\text{sign}} > A_{\text{random}} + \text{margin}, \tag{5}$$

$$\text{CE}(\theta_I) < \text{CE}(\theta_{\text{ref}}) \quad \text{on held-out data.} \tag{6}$$

Here A_{sign} measures agreement between the imported signs and held-out CE descent signs. The second check asks whether those signs matter for prediction, not just for a sign metric. A high alignment score without held-out CE improvement is treated as insufficient.

4.2 Train, Calibration, And Held-Out Separation

The corpus stream is split into training, calibration, and held-out regions. The training region accumulates G_D. The calibration region selects density, scale, or threshold parameters. The held-out region is not used by either step. This separation is necessary because TSSI can otherwise collapse into a corpus-specific table that looks useful only on the evidence used to build it. For the byte-level gates below, the main split is 300,000/100,000/100,000 tokens from `wiki_1mb.bin`.

4.3 Controls

Each control removes a different easy explanation.

Zero or uniform start. This baseline tests whether the imported artifact improves the reference cross-entropy at all.

Density-matched random ternary start. This baseline preserves sparsity and ternary occupancy while destroying corpus placement. Passing it means the result is not simply caused by putting enough nonzero signs into the readout or state summary.

Sign-shuffled start. This baseline preserves the empirical sign multiset while breaking the association between signs and coordinates. Passing it means the coordinate placement carries information.

Layer-summary controls. For live SSM states, `all`, `last`, and `mean` layer summaries test whether the signal depends on the full recurrent state structure or survives only as a weak pooled trace.

4.4 Record-Disjoint Rotations

The strongest leakage concern is that a small trace may let the compiler learn the flattened examples rather than a reusable state-coordinate statistic. Record-disjoint rotations address this by holding out whole trace records and rotating the held-out pair. A result that survives every rotation is not proof against all memorization, but it rules out the simplest explanation: one lucky held-out suffix or direct flattened-example reuse.

4.5 Cost Requirement

An end-to-end training claim requires:

$$R_{\mathrm{TSSI}} = h_{\mathrm{build}} + h_{\mathrm{realize}} + h_{\mathrm{verify}} + \kappa_{\mathrm{res}} \frac{H_{\mathrm{res}}}{H_0} \leq 0.90. \tag{7}$$

This inequality is a deployment requirement, not the definition of TSSI. The present paper focuses on whether the imported ternary artifact carries useful cross-entropy information in model coordinates.

4.6 Kill Criteria

TSSI should be rejected, or at least demoted, under any of the following conditions: held-out CE fails to beat matched random controls; sign alignment is high but prediction does not improve; sign-shuffled controls match the import; record-disjoint rotations collapse; live SSM coordinates lose the signal seen in symbolic proxies; or residual CE training exposes the imported signs as harmful. These are not secondary checks. They are part of the definition of a useful import.

5 Experiments

The experiments are organized as an evidence ladder. The first gate asks whether corpus moments can produce useful ternary signs in a simple byte-level feature map. The second asks whether the imported artifact remains useful after ordinary residual CE training begins. The third replaces symbolic features with stateful recurrent features. The fourth uses live Vulkan SSM hidden states. The last gate rotates held-out trace records to reduce the risk that the live result is a small-trace artifact.

5.1 Byte-Level Corpus Moment Gate

The initial byte-level gate used `wiki_1mb.bin` with 500,000 tokens and a 300,000/100,000/100,000 train/calibration/held-out split. The selected feature map was a bigram map, with density 0.05 and scale 2.4. The imported artifact reached held-out CE 2.949221092857901, compared with uniform CE 5.545177444479562 and random held-out CE mean 5.722178622339609. The weighted sign alignment was 0.9986051321029663, versus random mean 0.45805180817842484.

This is the first important separation. The imported signs are not merely nonzero ternary noise. They agree with held-out CE descent directions and reduce held-out CE in the same gate.

5.2 Residual Original-CE Pilot

The byte/bigram result could still be an initialization illusion: an artifact may look good before training, then lose its advantage once residual parameters start moving. To test this, a residual dense readout was trained on top of fixed imported or control logits under the same original CE target. After 1,000 residual steps, TSSI reached final CE 2.5494919515235193. The best control reached final CE 2.78477980855248, giving a same-budget CE advantage of 0.2352878570289607. The sign-shuffled control finished at 3.2129224560678225 and did not reach the imported target.

This pilot matters semantically because it tests the residual contract directly. The imported artifact does not merely lower an initial diagnostic number; it leaves the residual learner in a better CE state than zero, random, or sign-shuffled starts in this scope.

5.3 Stateful Proxy and Live SSM Trace

The next concern is that a byte bigram feature map may be too close to a direct local statistics table. A deterministic recurrent `ssm_state` proxy was therefore used as a stateful feature map. With density 0.01 and scale 1.6, two residual-pilot seeds preserved the advantage: seed 777 produced same-budget CE advantage 0.05614244344347741, and seed 419 produced 0.05664628606735311.

The stronger test used actual Vulkan SSM hidden states. The trace had shape $10 \times 6 \times 1 \times 1024 \times 128$ and, after materialization from `vulkan_fwd_h_cache`, all $7,864,320$ hidden-state values were nonzero. The best all-layer import used density 0.05 and scale 3.2, reaching held-out CE 5.346169927602639 versus random held-out CE mean 5.54442637453797. Weighted sign alignment

was 0.9970957040786743, versus random mean 0.5062233805656433. Under the residual pilot, the same-budget CE advantage was 0.19843119959677402.

The live trace is the current nucleus of the result. It moves TSSI from a symbolic corpus statistic into the coordinate system the model actually uses.

5.4 Record-Disjoint Anti-Memorization Gate

The live trace is small, so the anti-memorization gate is essential. The trace was split into records, and every two-record block was held out once under rotations $(8, 9)$, $(0, 1)$, $(2, 3)$, $(4, 5)$, and $(6, 7)$. All five all-layer rotations passed. The CE margin versus the best density-matched random minimum had min 0.189190727524743, mean 0.204293772528715, and max 0.238266453598485. Weighted sign alignment stayed at least 0.9929543137550354.

Last-layer and mean-layer summaries also passed all rotations, but with weaker mean margins: 0.0327855562301441 for last-layer summaries and 0.013581927197886 for mean-layer summaries. This supports the interpretation that the all-layer state carries the strongest corpus-directional statistic. The weaker summaries are useful controls, not the main result.

6 Analysis

The main result is not that a bigram feature map can predict bytes. That would be too weak and too close to a conventional count table. The meaningful result is that the same import idea survives a sequence of increasingly hostile reinterpretations: it survives held-out CE, sign-shuffled controls, residual training, a recurrent proxy, live SSM hidden-state coordinates, and record-disjoint rotations.

The live all-layer trace is the nucleus. The all-layer Vulkan trace is the strongest current evidence because it places the statistic in the actual recurrent state coordinates of the model. A byte-level or bigram gate can be useful as an identifying probe, but it does not by itself establish that TSSI is a model-native object. The live trace changes the status of the idea: the corpus-derived signs are no longer only symbolic features over token neighborhoods; they are useful over hidden states produced by the SSM path.

The controls remove simple explanations. The density-matched random controls remove the explanation that any sparse ternary pattern helps. The sign-shuffled controls remove the explanation that the sign balance alone is enough. The residual pilots remove the explanation that the import only wins before optimization begins. Record-disjoint rotations remove the simplest small-trace leakage story. None of these checks proves the mechanism at full scale, but together they make the positive signal hard to dismiss as formatting, density, or one lucky split.

The result is still narrower than a trained model. TSSI should be read as a partial commitment mechanism. It does not know the whole model. It does not decide every ternary coordinate. It does not claim that the imported artifact is a final parameter state. Its job is to move a subset of sign/zero decisions from iterative discovery into a corpus-statistic compiler, while leaving the residual CE problem alive. That is exactly why the zero state matters: the mechanism has a way to say "not enough evidence."

Why the idea is project-specific. The small ternary alphabet is not an incidental implementation detail. It is what makes the import question well-posed. For a real-valued model, the corpus would need to justify a magnitude, scale, and interaction pattern. For a ternary SSM, the compiler can ask a sharper first question: should this coordinate begin negative, zero, or positive under the original CE geometry? That makes TSSI different from generic data-dependent initialization. It is not only using data before training; it is using data to answer the discrete commitment question that this model family actually exposes.

7 Limitations

The current evidence is intentionally limited. These limitations are not decorative; they define what the paper can honestly claim.

Byte-level data. The gates use byte-level shards rather than a full BPE or unigram tokenizer pipeline. This makes the first experiments simple and repeatable, but it leaves open whether the same import signal holds under the tokenization and sequence statistics used by larger language-model training runs.

Small live trace. The live Vulkan SSM evidence comes from a 10-step batch-1 trace. The trace is valuable because it is real model state, but it is still small. Record-disjoint rotations reduce the easiest leakage concern, yet they do not replace a larger corpus-scale live-state validation.

Fixed feature maps. TSSI is only as good as the feature map in which corpus moments are accumulated. A weak feature map can miss useful structure; an overly expressive feature map can become too close to hidden retrieval. The live all-layer result is encouraging because it uses SSM states, but the feature-map question is still central.

Threshold selection. The current gates use density, scale, and calibration choices. These are reasonable for identifying whether the signal exists, but a mature TSSI system should derive or learn threshold rules that are stable across corpora, model sizes, and training stages.

No downstream capability result. The evidence is CE-centered. It shows that corpus-derived ternary commitments can reduce held-out CE and preserve residual CE advantage in the tested scopes. It does not yet show downstream task improvement, long-context capability transfer, or full training-run behavior.

Residual conflict remains possible. An imported sign can be locally useful and still interfere with later residual learning. The current residual pilots did not expose that harm in the tested scopes, but larger models may reveal interaction effects between imported commitments, SSM dynamics, and later optimizer updates.

8 Conclusion

This paper introduced TSSI, a corpus-to-ternary-artifact mechanism for state-space language-model training. The core idea is simple: if the model's parameter alphabet is ternary, then a corpus can sometimes answer part of the initial sign/zero decision problem before iterative optimization begins. The right import object is not raw text, not a retrieval table, and not a teacher model. It is a sufficient statistic for original cross-entropy descent in a fixed model-coordinate feature map.

The current evidence supports the mechanism. Corpus-derived ternary commitments beat density-matched random and sign-shuffled controls, survive residual original-CE training, remain useful in a recurrent state proxy, and carry signal over live Vulkan SSM hidden states. Record-disjoint rotations further show that the strongest all-layer live-state result is not explained by a single held-out suffix or trivial flattened-example reuse.

The next experiment is direct: build a model-native TSSI path that accumulates the same kind of corpus moments without relying on an offline trace export, then compare imported, zero, random, and sign-shuffled starts under residual original-CE training at larger corpus and model scale. If that holds, TSSI becomes more than an identifying gate. It becomes a new way to make the corpus participate in constructing the initial ternary model state.

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